

Routing Protocol in Wireless Sensor Network (Wsns)

Abstract

Wireless Sensor Networks (WSNs) have drawn a lot of interest because of its many uses, which include surveillance, military operations, smartphones, healthcare monitoring, and disaster relief. Usually, these networks are made up of numerous sensor nodes placed in hostile and unsupervised areas. Energy efficiency is a major challenge because these nodes run on a limited amount of battery power. Clustering, which groups nodes to maximize communication, is a commonly used strategy to lower energy use. For this, hierarchical clustering systems are frequently employed, especially cluster-based techniques.

Sensor nodes are grouped into clusters using cluster-based techniques, and one node is chosen to serve as the Cluster Head (CH) in order to oversee communication. The network is separated into virtual grids in grid-based systems, which are typically controlled by the base station. This paper highlights and discusses the design challenges for cluster-based schemes, the important cluster formation parameters, and classification of hierarchical clustering protocols. Moreover, existing cluster-based techniques are evaluated by considering certain parameters to help users in selecting appropriate technique. Furthermore, a detailed summary of these protocols is presented with their advantages, disadvantages, and applicability in particular cases.

Key Words: Data Clustering, K-Means Algorithm, Data Mining, Cluster Analysis, Centroid, Scalability, Real-world Applications, Iterations, Convergence, Feature Selection.

1. Introduction

Wireless Sensor Networks (WSNs) consist of a large number of sensor nodes used to monitor physical or environmental conditions such as temperature, sound, vibrations, or motion. These nodes detect, process, and forward collected data to a base station or sink for application-specific analysis [1]. Since sensor nodes have limited energy resources and computational capabilities, an efficient data transmission is required to extend the lifetime of the network

The direct communication between nodes and the sink is usually not feasible because transmission with high power is needed that exhausts the sensor nodes energy reserve in a short time. Instead, nodes cooperate in a multi-hop peer-to-peer communication manner to reach the sink using message forwarding through intermediate nodes. It balances energy and prolongs the network's function at large deployment areas. Routing in Wireless Sensor Networks (WSNs) presents significantly greater challenges than in traditional wired or wireless networks due to the resource-constrained and dynamic nature of these systems. A primary consideration in designing routing protocols for WSNs is energy efficiency, as sensor nodes are often battery-powered and deployed in inaccessible areas where recharging or replacement is impractical. This limitation requires protocols to optimize energy consumption to prolong network lifespan [1].

Additionally, lack of centralized control means that routing decisions must be made locally by individual nodes, relying on distributed, lightweight algorithms. As WSNs can scale to hundreds or thousands of nodes, protocols must maintain high scalability without depending on global knowledge of the network.

Node failures—whether due to hardware malfunctions, environmental conditions, or energy depletion—necessitate robust and adaptive routing mechanisms that can rapidly reconfigure paths and maintain communication. The dynamic topology of WSNs, influenced by node mobility and environmental factors, further reinforces the need for flexibility in routing strategies.

Moreover, many WSNs consist of heterogeneous nodes with varying capabilities, making protocol adaptability essential. In cases where mobility is involved—such as mobile sinks or mobile sensor nodes—routing protocols must ensure seamless connectivity. Lastly, Quality of Service (QoS) demands in time-sensitive or critical applications require a balance between low latency, high delivery rates, and energy conservation, adding complexity to the routing process [1].

2. Literature Review

Over the years, a wide range of routing protocols have been developed for Wireless Sensor Networks (WSNs), each tailored to address specific network constraints such as limited energy, scalability, and communication reliability. Early routing techniques like Flooding and Gossiping offered simplicity and broad applicability but suffered from issues like data redundancy, energy inefficiency, and implosion effects due to uncontrolled message propagation. These limitations led to the development of more structured protocols that optimize routing decisions based on node capabilities and application demands.

Significant advancements have been made with attribute-based, hierarchical, and location-based routing schemes. Protocols such as SPIN and Directed Diffusion employ attribute-based mechanisms to reduce redundant transmissions by negotiating data propagation using meta-information. Hierarchical approaches like LEACH and PEGASIS introduced clustering mechanisms to distribute energy load by assigning roles to sensor nodes (e.g., cluster heads), thereby enhancing network lifetime and reducing communication overhead. Location-based protocols leverage geographic position data to route packets efficiently, although they often depend on additional hardware like GPS, increasing deployment costs.

Moreover, recent research emphasizes the importance of energy-efficient and QoS-aware routing, especially for mission-critical applications. Multipath routing strategies have also been proposed to ensure fault tolerance and load balancing by maintaining several disjoint paths between source and sink. Notably, the paper also highlights contributions from Spanish research institutions, which have focused on enhancing routing reliability and robustness through innovative algorithms and optimization techniques. This evolving body of work continues to shape the design of resilient, adaptable, and energy-conscious routing protocols suited for modern WSN applications.

3. Cluster-Based Organization in Wireless Sensor Networks (WSNs)

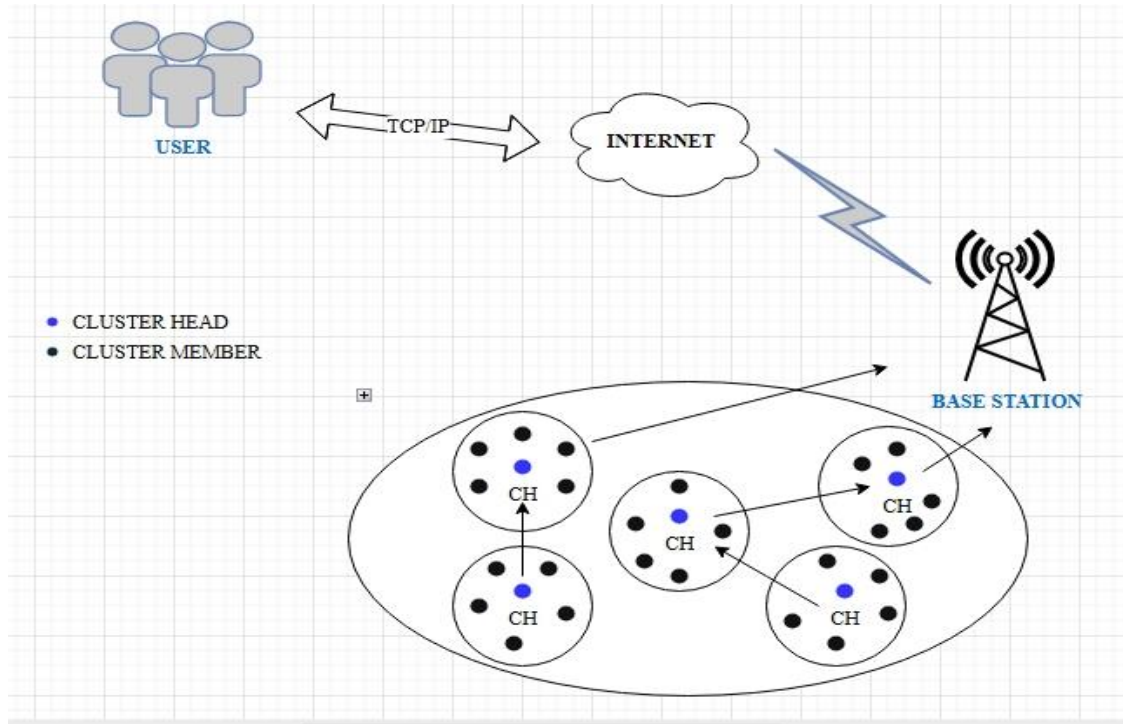


Fig (1): Cluster Based Organization in Wireless Sensor Network [3,36]

Given the limited resources available in Wireless Sensor Networks (WSN), direct communication between sensor nodes and the Base Station (BS), or the multihop communication of sensor nodes towards the BS, is impractical due to high energy consumption, which leads to the premature failure of sensor nodes. Direct or single-tier communication is not viable for large-scale networks since WSNs cannot accommodate long-distance communication. This method has several drawbacks, including excessive energy usage, data redundancy (where nearby sensor nodes transmit data with minimal variation), and the rapid depletion of energy in the most distant nodes. To address these challenges, a two-tier communication model utilizing a hierarchical approach is implemented, wherein nodes are organized into clusters. The leader node, known as the cluster head (CH), is tasked with aggregating data and subsequently relaying it to the BS. This hierarchical network structure typically establishes a two-level hierarchy, with cluster heads positioned at the upper level and member nodes at the lower level. The lower-level nodes regularly transmit data to their corresponding CH, which then consolidates this information and forwards it to the BS. The CH expends more energy than the member nodes, as it is consistently transmitting data over extended distances. Furthermore, after a certain number of rounds, the selected CH may become ineffective or fail due to high energy consumption. To promote load balancing among sensor nodes, the function of the Cluster Head (CH) is periodically adjusted to optimize energy consumption [3, 36]. Communication within a cluster occurs in a single-hop manner (intracluster), while intercluster communication is conducted through multiple hops. Among the hierarchical techniques, cluster-based and grid-based methods are the most prevalent [11].

4. Energy-Efficient Hierarchical Clustering Techniques in WSNs

One of the major challenges in Wireless Sensor Networks (WSNs) is managing energy efficiently, especially since sensor nodes typically run on limited battery power and are often placed in remote or difficult-to-reach areas. Because replacing or recharging these batteries

isn't always practical, extending the network's operational life becomes a top priority. To tackle this, researchers have developed various hierarchical clustering techniques. These methods group sensor nodes into clusters, each led by a node known as the Cluster Head (CH). The CH takes on the responsibility of collecting and sending data to the base station, which helps reduce the energy spent by individual nodes on long-range communication.

In the study, hierarchical clustering techniques are generally divided into two categories:

Cluster-based methods involve organizing sensor nodes into clusters, where one node—the CH—manages data gathering and transmission. Some well-known protocols in this category include LEACH, LEACH-C, and CHEF. These protocols typically aim to improve the process of choosing cluster heads, often using probabilistic models, energy levels, or fuzzy logic. They also look at how to best form clusters and evenly spread the communication load across the network to avoid overburdening any one node.

4.1. Low Energy Adaptive Clustering Hierarchy (LEACH)

LEACH, or Low Energy Adaptive Clustering Hierarchy, was one of the earliest and most influential routing protocols designed specifically for energy efficiency in Wireless Sensor Networks (WSNs). Introduced by Heinzelman et al. [15], it remains a widely referenced benchmark for energy-aware routing in sensor networks.

The main concept behind LEACH is to rotate the role of the Cluster Head (CH) among all sensor nodes to evenly distribute energy usage across the network. This rotation helps prevent any single node from depleting its battery too quickly due to repeated data transmissions.

LEACH works in two main phases: the setup phase and the steady-state phase. During the setup phase, each sensor node determines whether it should become a CH for the current round. This decision is based on a predefined probability and the number of times the node has previously served as a CH. Each node generates a random number between 0 and 1. If this number is less than a calculated threshold, the node elects itself as a CH. The threshold is determined using the following formula:

$$T(n) = \begin{cases} \frac{P}{1 - P \times (r \bmod \frac{1}{P})}, & \text{if } n \in G \\ 0, & \text{otherwise} \end{cases}$$

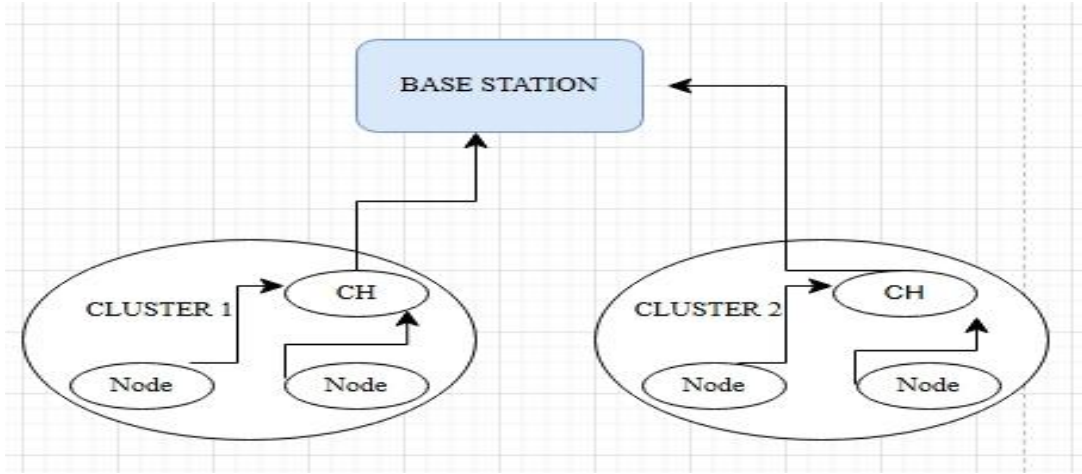


Fig (2): LEACH Algorithm [15]

Here:

- P is the desired percentage of CHs in the network,
- r is the current round number,
- G is the set of nodes that haven't been CHs in the past 1/P rounds.

This ensures fair CH rotation over time. Once CHs are selected, the steady-state phase begins, where data is collected from member nodes, aggregated by the CH, and transmitted to the base station.

4.2. Low Energy Adaptive Clustering Hierarchy - Centralized (LEACH-C)

LEACH-C is an enhanced version of the original LEACH protocol, with a key difference: instead of letting nodes decide on their own cluster heads (CHs), the base station (BS) takes control. It gathers data on each node's energy level and location, then centrally decides the optimal number of CHs and how the network should be clustered.

The BS selects nodes with above-average energy to serve as CHs and ensures an even distribution of clusters. To do this efficiently, it may split the network into subclusters and use algorithms like minimum spanning tree to determine the best routing paths.

While this centralized method leads to better-balanced clusters and more stable CH selection, it comes at a cost. The BS must frequently collect updates, and the communication between nodes and the BS increases, leading to higher energy consumption and overhead—especially during CH reselection.

4.3. Cluster Head Election Using Fuzzy Logic (CHEF)

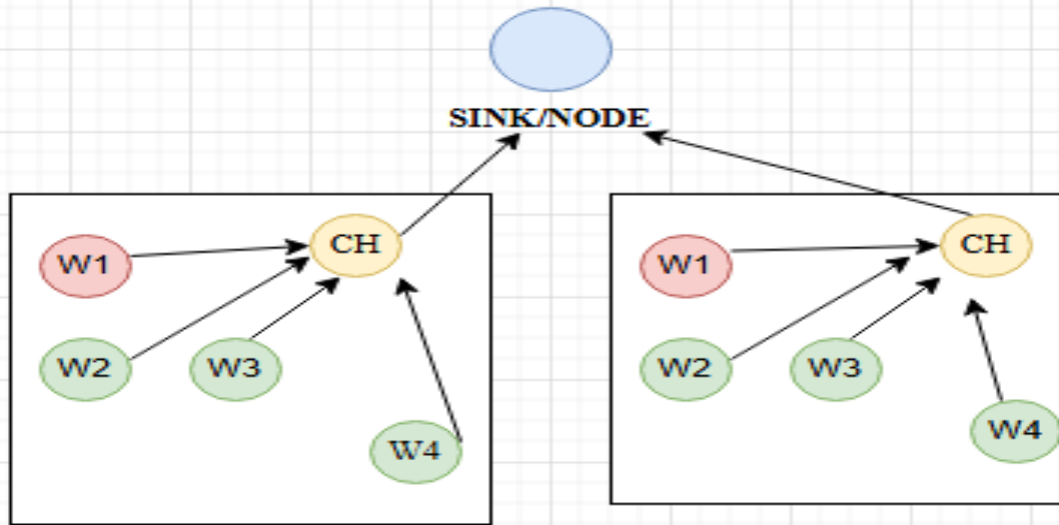


Fig (3): CHEF Algorithm [19]

The CHEF protocol, introduced by the authors in [19], uses fuzzy logic to make the process of choosing cluster heads (CHs) more intelligent. Initially, a few nodes are randomly picked as tentative CHs. Each of these candidates then evaluates two key parameters: its energy level and its local distance (which is the total distance to neighboring nodes).

Using fuzzy “if-then” rules, each candidate calculates a chance value that reflects how suitable it is to serve as a CH. The node with the highest chance value becomes the actual CH and broadcasts its status so that nearby nodes can join its cluster.

While CHEF enhances the overall lifetime of the network by making smarter CH selections, it also introduces some downsides. Since the CH selection process is repeated frequently across the entire network, it leads to extra message exchanges, creating unnecessary traffic and higher energy consumption.

4.4. K-Means Algorithm

The K-Means algorithm is one of the most widely used unsupervised machine learning techniques for clustering, where the goal is to partition a dataset into K *distinct, non-overlapping groups* based on data similarity. It operates by initializing K *centroids* randomly and then iteratively refining them by assigning each data point to the nearest centroid and updating the centroids as the mean of the assigned points. This process continues until the centroids stabilize and no longer change significantly, indicating convergence [1].

One of the key reasons for K-Means' popularity is its computational efficiency and simplicity, which makes it suitable for large datasets. However, the performance of K-Means is sensitive to the initial selection of centroids and the pre-defined value of K . Additionally, the algorithm assumes that clusters are spherical and of similar size, which may not hold true in real-world applications.

Despite its limitations, K-Means remains effective for many practical applications such as market segmentation, image compression, and pattern recognition, where the underlying data naturally forms clusters. Numerous enhancements and variations of the original algorithm have been proposed to improve its accuracy, scalability, and robustness in different scenarios [1].

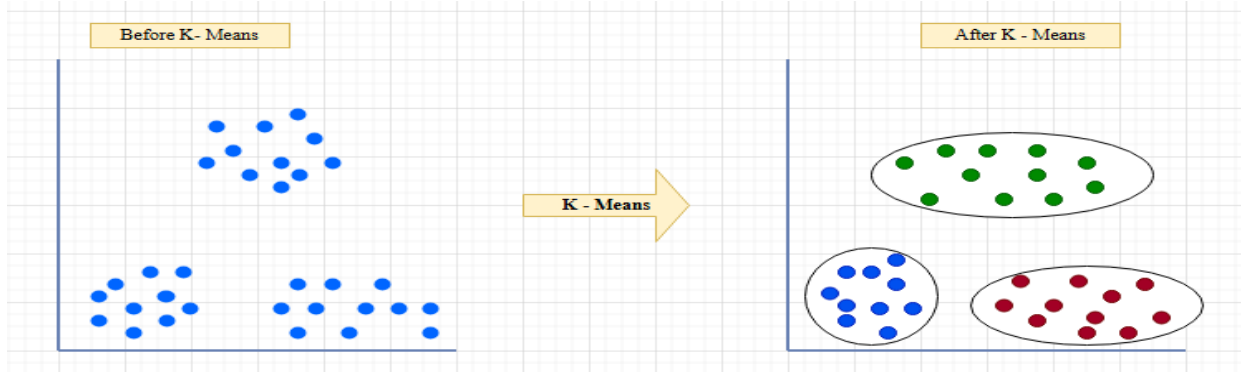


Fig (4): K-Mean Algorithm

4.5. Cluster-Based Centralized Weighted Management (CCWM)

CCWM is a clustering technique designed specifically for Wireless Sensor Networks (WSNs) to enhance energy efficiency and prolong network lifetime. In this approach, the network is divided into clusters, each managed by a Cluster Head (CH), which is selected based on a weighted combination of parameters such as residual energy, distance to the base station, and node density. The "centralized" aspect refers to the fact that the Base Station (BS) performs the CH selection and clustering process, ensuring optimal decisions with a global view of the network. By balancing the load among sensor nodes and reducing long-range communication, CCWM effectively minimizes energy consumption and improves scalability and reliability in WSNs [1].

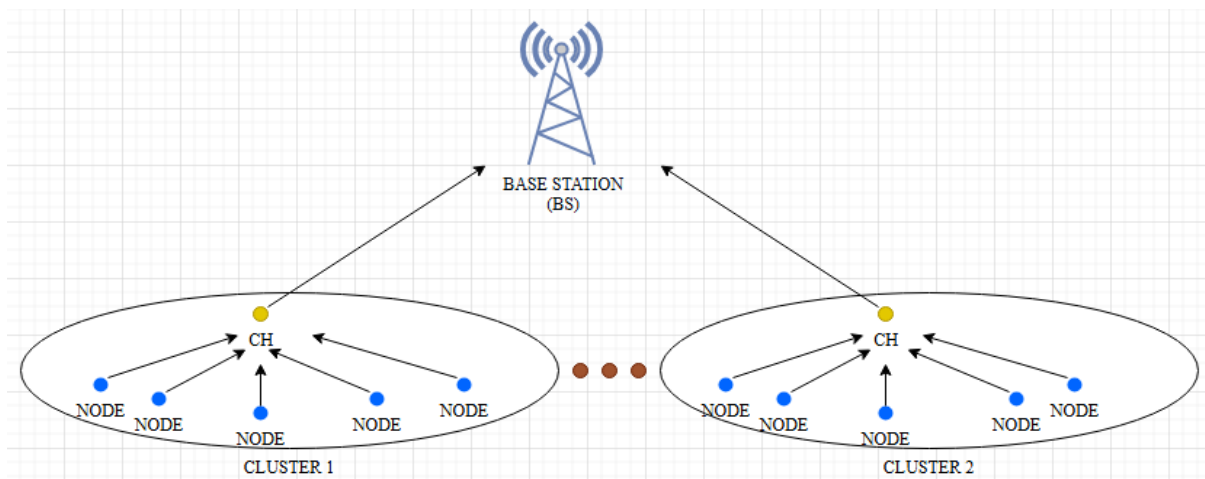


Fig (5): CCWM Algorithm

5. Current Challenges and Future Directions

This paper explores various hierarchical clustering algorithms, focusing on specific parameters. A comprehensive comparison of existing algorithms is presented, highlighting the need for further research, particularly in integrating these algorithms with emerging technologies like the Internet of Things (IoT) and vehicular ad hoc networks (VANETs). A major challenge identified is the security of wireless sensor networks (WSNs), which typically operate in open environments, making them vulnerable. Traditional cryptographic methods cannot be used in WSNs due to resource limitations, necessitating the development of lightweight security mechanisms that minimize energy consumption while ensuring secure data transmission.

In the context of VANETs, sensors deployed on vehicles can help monitor events and ensure road safety. However, the high mobility of vehicles presents a challenge for efficient data aggregation using sensors. Similarly, as IoT connects a wide range of devices, it will generate vast amounts of data, posing challenges in terms of connectivity, retrieval, storage, and security. This massive volume of data, known as big data, requires advanced deep learning techniques for effective query processing and information retrieval.

Multimedia sensors, commonly used for image capture in targeted areas or upon triggering specific events, play a crucial role in surveillance and security. When deploying sensors for such applications, it is essential to consider factors like delay and fault tolerance to ensure their effectiveness and reliability.

The cluster-based hierarchical energy-efficient protocols in a table format:

Protocol	CH Selection Approach	Type	Advantages	Disadvantages
LEACH	Probabilistic	Distributed	(i) Nodes share load to some extent (ii) TDMA avoids collisions (iii) Allocated time slots reduce energy consumption	(i) Single-hop inter-cluster communication (ii) Energy holes (iii) CH selection without energy consideration (iv) Extra overhead from dynamic clustering
LEACH-C	Probabilistic	Centralized	(i) Global network view (ii) Even load distribution (iii) Energy-efficient routes	(i) Network overhead (ii) Probabilistic CH selection (iii) Resource-expensive reselection
CCWM	Non probabilistic	Distributed	(i) Improves routing & network lifetime (ii) Suitable for small networks	(i) Non-optimized CH election (ii) Increases

Protocol	CH Selection Approach	Type	Advantages	Disadvantages
				network overhead (iii) Not for large networks
K-means	Probabilistic/Centroid-based	Distributed	(i) Simple approach (ii) Improved network lifetime	(i) Periodic cluster reformation (ii) Inefficient load distribution (iii) Reselection based on centroid distance
CHEF	Probabilistic/Nonprobabilistic	Distributed	(i) Optimal CH selection considering the entire network	(i) High message & communication overhead (ii) Expensive CH election process
UCS	Nonprobabilistic/Probabilistic	Distributed	(i) Balanced energy consumption (ii) Suitable for homogeneous networks	(i) Varying node number with random deployment (ii) Optimal CH number in multilayer approach
NUDND	Nonprobabilistic	Distributed	(i) Load balancing (ii) Resolves energy hole problem (iii) Works well with predefined node positions	(i) Not suitable for random deployment (ii) Not for mobile nodes or large-scale networks
EADC	Nonprobabilistic	Distributed	(i) Load balancing (ii) Addresses imbalance in energy consumption	(i) Redundant messages increase energy consumption (ii) Uneven clustering in random deployments

Protocol	CH Selection Approach	Type	Advantages	Disadvantages
LEACH-MAC	Probabilistic	Distributed	(i) Addresses randomness in CH count (ii) Nodes share load (iii) Energy-efficient	(i) Random CH selection (ii) Single-hop inter-cluster communication (iii) Probabilistic CH selection without energy consideration
EADUC	Nonprobabilistic	Distributed	(i) Solves energy hole problem (ii) Maximizes network lifetime (iii) Unequal clustering strategy	(i) Redundant messages increase energy consumption (ii) Extra energy use affects performance

6. Conclusion

Wireless Sensor Networks (WSNs) are gaining increasing attention due to their affordability, compact size, and the ability to deploy battery-powered sensors in harsh, geographically challenging areas. These sensor nodes are integrated with other emerging technologies like the Internet of Things (IoT), mobile phones, and IEEE 802.11, positioning WSN as a key technology for the 21st century. However, operating in hostile environments with limited battery power makes WSNs a complex area for research. This paper aims to provide a comparative analysis of existing sensor node deployment strategies and energy-efficient clustering protocols, examining their relative advantages and limitations.

Since WSNs are resource-constrained and often operate in remote, unattended environments, replacing or recharging batteries is not feasible. To make the most of their limited energy, several energy-efficient protocols have been proposed. Among these, cluster-based protocols have garnered significant attention due to their ability to extend the network's overall lifespan. Most of these methods focus on hierarchical clustering, where nodes are grouped into clusters, and the cluster head forwards the data to the base station (BS). This paper critically evaluates the state-of-the-art cluster-based energy-efficient techniques in WSNs, considering various parameters such as cluster formation metrics, energy consumption, and network lifetime.

Additionally, the paper discusses the design challenges and research hurdles associated with hierarchical approaches. Based on the evaluation criteria, a comparative analysis is presented to guide the selection of the most appropriate technique for specific needs. The paper also highlights the importance and limitations of both clustering techniques, providing insights into their suitability for particular operational environments.

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